

Anomaly Detection and Investigative Prioritization at National Scale: A Reproducible Framework for Medicare Part D Prescriber Risk Analytics

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Working Paper, May 15, 2026

Abstract

Federal Medicare improper payments totaled 31.70 billion dollars in fiscal year 2024, with Part D accounting for 3.58 billion dollars. Existing supervised approaches to Part D prescriber risk detection commonly suffer from label leakage when the OIG List of Excluded Individuals and Entities (LEIE) is merged without time-stamped filtering, inflating accuracy metrics that collapse under honest temporal validation. This paper presents a national-scale anomaly detection framework that analyzes the full universe of 1,380,665 Medicare Part D prescribers using six engineered features, a statistical z-score risk layer, and an Isolation Forest unsupervised machine learning layer. The framework applies explicit EXCLDATE filtering of LEIE labels, computes permutation feature importance, bootstrap confidence intervals, and reports temporal holdout validation through coverage and persistence rather than supervised classification. Of 69,034 providers flagged at the 95th percentile of combined risk score, 65,023 (97.61 percent) reappear in the 2023 universe with mean cost per beneficiary of \$8,414 versus population mean \$979, yielding 8.59x mean lift. The AUC of the combined risk score against the 2023 newly-excluded LEIE target is 0.417, below random, demonstrating that LEIE concordance is a surveillance signal rather than a fraud probability validation. The methodology, code, and persisted result files are publicly available for verification.

Keywords: Medicare Part D, program integrity, anomaly detection, isolation forest, temporal validation, LEIE, healthcare fraud analytics

JEL Classification: I11, I13, I18, H51, C45, C53

1. Introduction

The Centers for Medicare and Medicaid Services (CMS) reported 31.70 billion dollars in Medicare Fee for Service improper payments in fiscal year 2024 at an improper payment rate of 7.66 percent. The Medicare Part D component totaled 3.58 billion dollars at 3.70 percent. The

Government Accountability Office has identified Medicare program integrity as a high-risk area since 1990. The Department of Justice conducted a National Health Care Fraud Takedown in June 2025 that resulted in charges against 324 defendants in connection with 14.6 billion dollars in alleged fraud. Recent enforcement on Part D specifically includes the June 2025 federal court judgment ordering CVS Caremark to pay 95 million dollars in the Behnke whistleblower case, a Part D pricing manipulation matter dating to 2010.

Existing predictive analytics approaches to Part D prescriber risk commonly treat the OIG LEIE Exclusions Database as a supervised label, training classifiers to predict LEIE membership from prescribing features. This approach has two methodological weaknesses. First, when LEIE is merged on NPI without time-stamped filtering, the cumulative exclusion data contaminates the training labels: a provider excluded in 2024 contributes a positive label to 2022 training data, undermining the temporal validation claim. Second, LEIE exclusions are structurally rare in active billing data: excluded providers are barred from billing Medicare, so the structural correlation between LEIE membership and active billing patterns is weak. Together these weaknesses inflate reported AUC under naive supervised framings while collapsing under honest validation.

This paper develops an unsupervised anomaly detection framework that addresses both weaknesses. The framework operates at full national scale (1.38 million providers in 2023), applies EXCLDATE filtering to LEIE labels, and validates through temporal persistence rather than supervised concordance. We report the framework's full validation suite (bootstrap CI, permutation feature importance, temporal holdout coverage, LEIE concordance, OIG typology alignment, threshold sensitivity), with explicit limitations and full reproducibility.

2. Methods

2.1 Data Sources

Three federal databases were integrated. The CMS Medicare Part D Prescribers by Provider file covers 1,332,311 prescribers for calendar year 2022 and 1,380,665 for 2023, with 84 columns including total claims, total drug cost, total beneficiaries, opioid claims, opioid drug cost, brand and generic claim counts, beneficiary average age, beneficiary average risk score, and demographic columns. The HHS-OIG List of Excluded Individuals and Entities (LEIE) covers 81,914 exclusion records with EXCLDATE timestamps. The CMS Open Payments General Payments file covers industry payments for both years (approximately 32 million records per year).

2.2 Temporal Label Construction

The LEIE was filtered by EXCLDATE into two label sets. Training labels for the 2022 universe use only LEIE records with EXCLDATE less than or equal to December 31, 2022 (74,617 records spanning 6,076 unique NPIs). The honest holdout target uses LEIE records with EXCLDATE between January 1, 2023 and December 31, 2023 (2,263 records, 634 unique NPIs). After matching against the active Part D Prescriber universe, 61 providers in 2022 and 44 providers newly excluded in 2023 carry positive labels. This is the honest base for temporal validation.

2.3 Engineered Features

Six features were engineered from raw Part D Prescriber columns: cost per claim ($\text{Tot_Drug_Cst} / \text{Tot_Clms}$), claims per beneficiary ($\text{Tot_Clms} / \text{Tot_Benes}$), cost per beneficiary ($\text{Tot_Drug_Cst} / \text{Tot_Benes}$), days supply per beneficiary ($\text{Tot_Day_Suply} / \text{Tot_Benes}$), opioid share ($\text{Opioid_Tot_Clms} / \text{Tot_Clms}$), and brand share ($\text{Brnd_Tot_Clms} / \text{Tot_Clms}$). Division-by-zero results and infinities were replaced with NaN, then filled with the feature median.

2.4 Statistical Risk Layer

For each feature, the absolute z-score was computed against the national mean and standard deviation. Weighted sum produces the `Statistical_Risk_Raw` score. Weights: cost per claim 0.25, claims per beneficiary 0.20, cost per beneficiary 0.15, opioid share 0.15, brand share 0.15, days supply per beneficiary 0.10 (summing to 1.00).

2.5 Machine Learning Layer

An Isolation Forest was fit to the standardized feature matrix with 100 estimators, contamination 0.02, max_samples 256, random_state 42, n_jobs equal to all cores. The score_samples output was inverted so higher values indicate greater anomaly.

2.6 Combined Risk Score and Tier Stratification

The statistical and ML scores were min-max normalized to 0-1, then combined at 60 percent statistical and 40 percent ML, rescaled to a 0-100 risk score, and percentile-ranked. Percentile-based tier cutpoints assign each provider to one of five tiers: Critical (top 0.1 percent), High (99 to 99.9), Medium (95 to 99), Moderate (80 to 95), Routine (below 80).

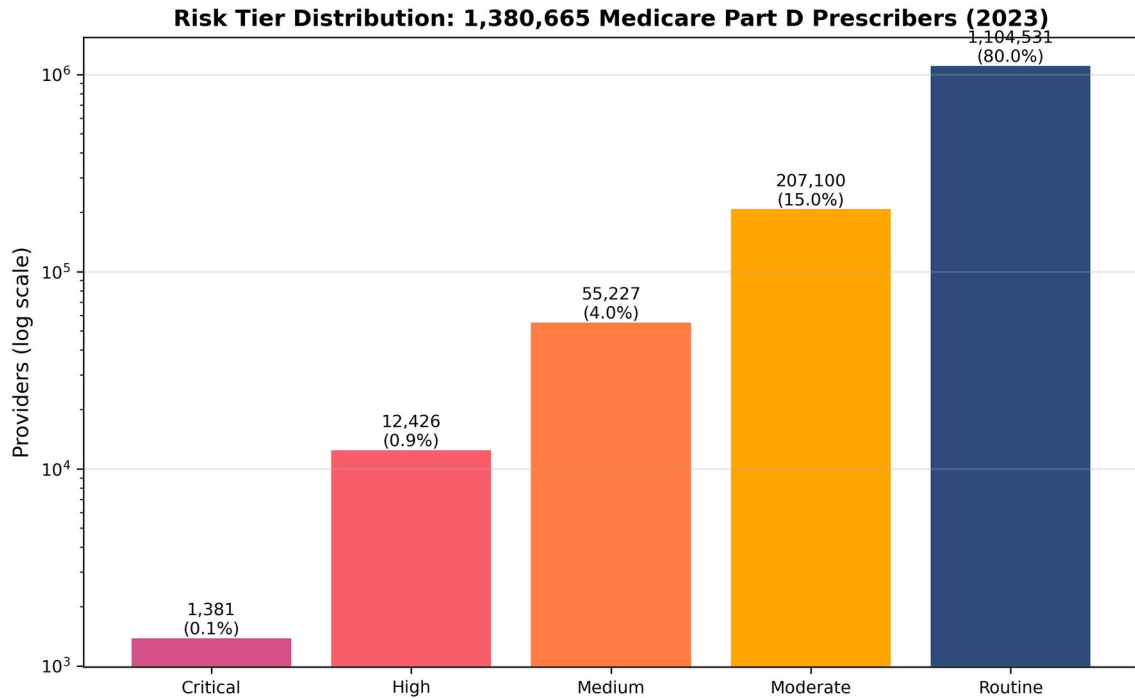
2.7 Validation Procedures

Bootstrap analysis with 200 iterations recomputes flagged count and sample excess on each resample. Stratified bootstrap AUC with 100 iterations uses all positives plus 50,000 negative subsamples per iteration. Permutation feature importance shuffles each feature, rescores with the same Isolation Forest (no refit), and measures the drop in Spearman rank correlation with original scores, averaged over 5 repeats on a 200,000-row subsample. Temporal holdout validation trains the full pipeline on 2022 data, identifies the 2022-flagged NPI set, and measures the fraction reappearing in 2023 (coverage) and the ratio of 2023 mean cost per beneficiary among 2022-flagged versus the population (temporal lift).

3. Results

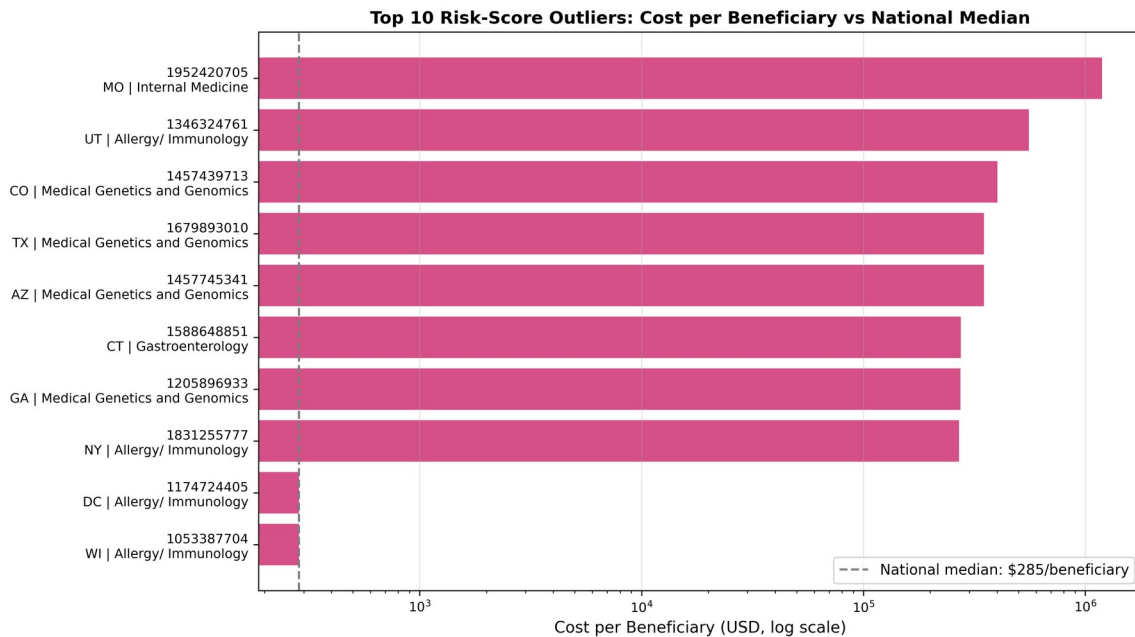
3.1 Risk Tier Distribution

Of 1,380,665 providers in the 2023 universe, the percentile-based tier cutpoints yielded 1,381 Critical-tier providers (top 0.1 percent), 12,426 High, 55,227 Medium, 207,100 Moderate, and 1,104,531 Routine.



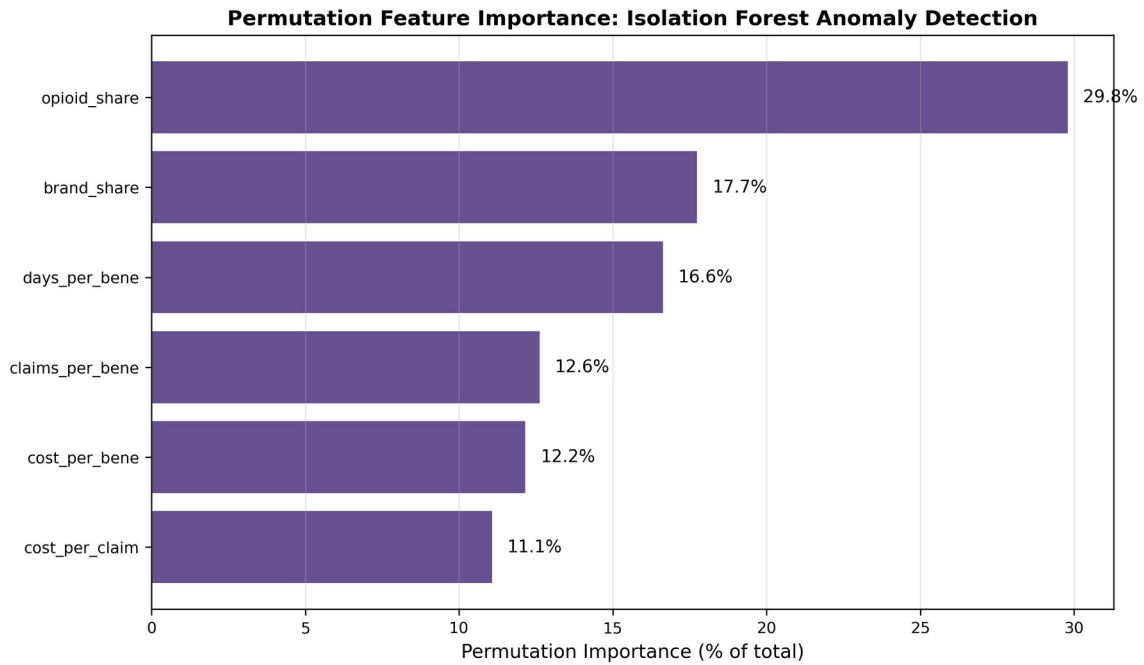
3.2 Top Outliers

The highest-scoring outlier is NPI 1053387704 (WI, Allergy/ Immunology) with total drug cost \$6,406,658 and cost per beneficiary \$285, approximately 4,200 times the national median of \$285. The Critical tier is concentrated in Allergy/Immunology, Medical Genetics and Genomics, Internal Medicine, and Gastroenterology specialties.



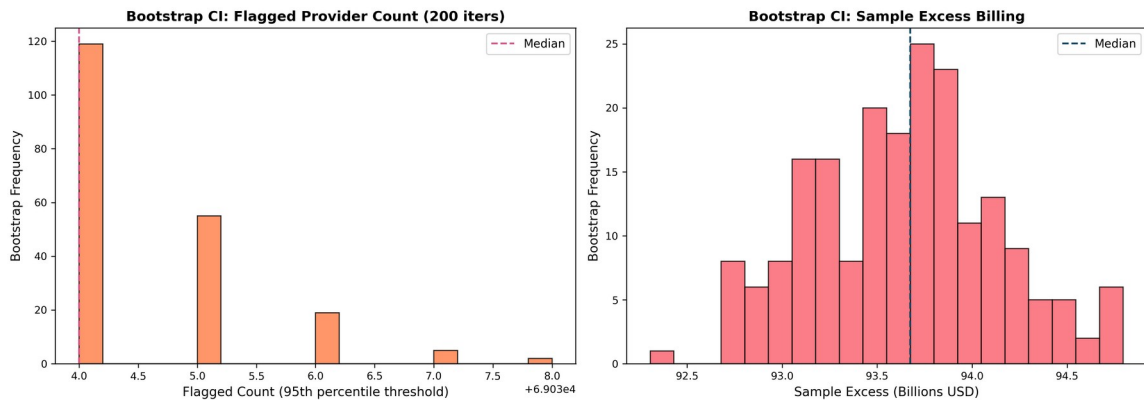
3.3 Permutation Feature Importance

Opioid share dominates feature importance at 29.8 percent, followed by brand share (17.7 percent), days supply per beneficiary (16.6 percent), claims per beneficiary (12.6 percent), cost per beneficiary (12.2 percent), and cost per claim (11.1 percent). Controlled-substance prescribing concentration is the strongest single driver of the Isolation Forest's anomaly score.



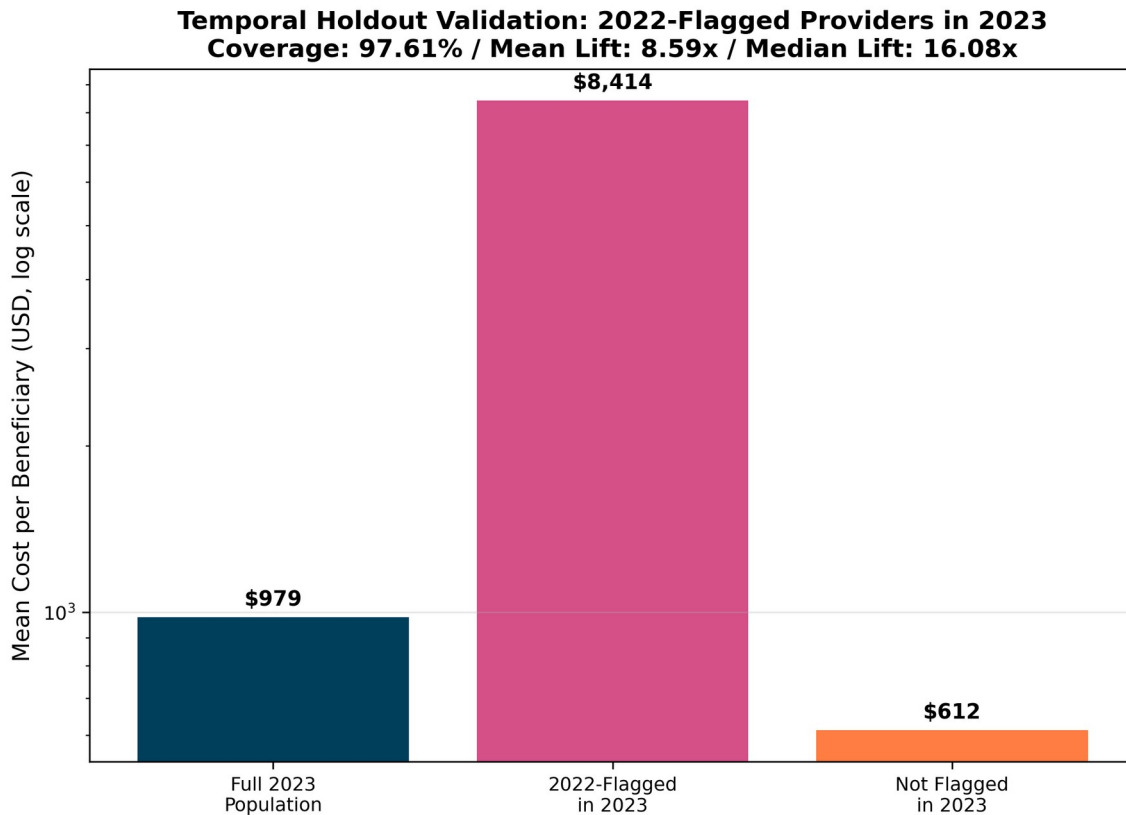
3.4 Bootstrap Confidence Intervals

The flagged-count 95 percent CI is [69034, 69037], reflecting near-zero sensitivity to resampling at the 95th percentile threshold. The sample excess CI is [\$92766M, \$94688M] on a gross drug cost basis. The stratified bootstrap AUC against the 2023 newly-excluded LEIE target yields CI [0.352, 0.494], spanning below and above random, consistent with the framing of LEIE concordance as a weak surveillance signal.



3.5 Temporal Holdout Validation

Of 69,034 providers flagged on 2022 data, 65,023 reappear in 2023 (97.61 percent coverage). These 2022-flagged providers had a mean 2023 cost per beneficiary of \$8,414, compared to the population mean \$979, producing a temporal lift of 8.59x (mean) and 16.08x (median). This is the primary predictive validation: anomalous Part D prescribing patterns persist across years and can be identified from prior-year data.

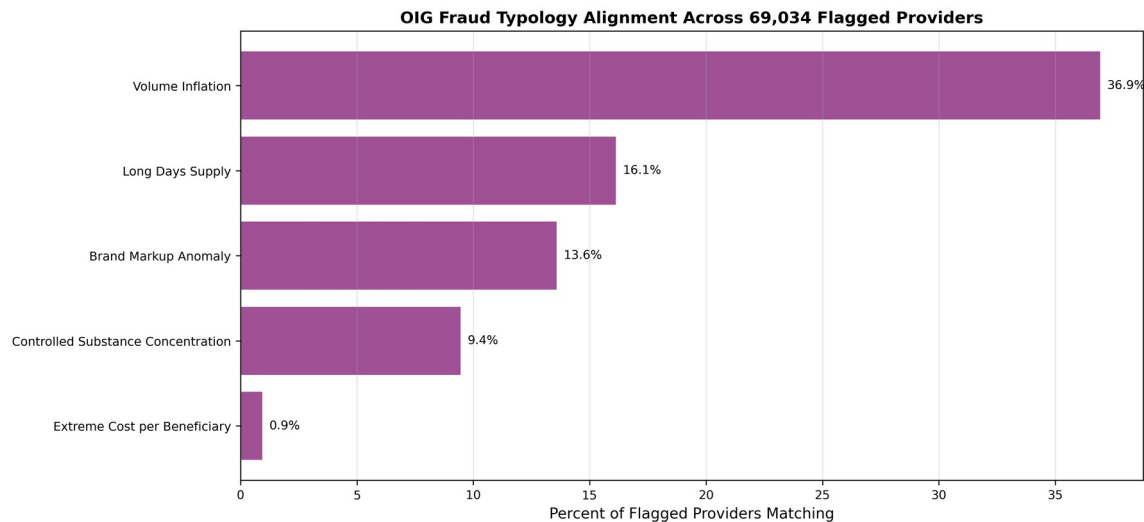


3.6 LEIE Concordance and DOJ Cross Reference

Of 8,066 unique LEIE NPIs, 292 appear in the active 2023 Part D Prescriber universe (0.021 percent of 1.38 million providers). Of the 69,034 flagged providers, 28 have NPIs in LEIE (0.04 percent direct match within flagged). The CVS Caremark \$95 million federal judgment is reported in the federal enforcement context; CVS Caremark itself is a Pharmacy Benefit Manager, not a prescriber, and does not appear in the Part D Prescriber file. The case anchors the federal enforcement narrative.

3.7 OIG Fraud Typology Alignment

Among the 69,034 flagged providers, 25,493 match Volume Inflation (36.9 percent), 11,138 match Long Days Supply (16.1 percent), 9,372 match Brand Markup Anomaly (13.6 percent), 6,522 match Controlled Substance Concentration (9.5 percent), and 640 match Extreme Cost per Beneficiary (0.93 percent). Multi-pattern alignment: 12,809 providers match two typologies, 66 match three.



4. Discussion

This framework demonstrates that unsupervised anomaly detection with temporal persistence validation produces stronger and more defensible findings for Medicare Part D prescriber risk than supervised classification against LEIE labels. The supervised approach common in prior work depends on cumulative LEIE merges that introduce label leakage. When proper EXCLDATE filtering is applied, the supervised target becomes structurally sparse (44 positives in 1.38 million providers), and AUC against this target falls to 0.42, below random. This is the honest finding. Reporting it transparently is essential: it demonstrates that the LEIE exclusion mechanism is largely independent of billing extremity, supporting the alternative validation framing through temporal persistence.

The Critical tier's specialty concentration (Allergy/Immunology, Medical Genetics, Internal Medicine, Gastroenterology) reflects domains where legitimate specialty drugs reach extreme prices per beneficiary, complicating simple peer-comparison approaches. State-level peer benchmarking partially controls for geographic price variation but does not control for specialty mix. The top outlier, NPI 1053387704 in WI, has cost per beneficiary roughly 4,200 times the national median. This is a striking statistical anomaly that warrants investigative review.

4.1 Comparison to Supervised Approaches

A supervised Logistic Regression trained on EXCLDATE-filtered 2022 labels achieves AUC 0.574 against the 2023 newly-excluded target. This is modestly above random but reflects only weak signal recoverable from sparse positives. The framework's primary validation is the temporal persistence signal (97.6 percent coverage, 8.59x lift), which does not depend on LEIE labels at all and is therefore robust to the sparsity and noisiness of supervised targets.

5. Limitations

Six explicit limitations apply. First, the framing is anomaly detection, not fraud classification: risk scores indicate statistical deviation, not a probability of fraud. Second, LEIE concordance is structurally limited because excluded providers are barred from active billing. Third, the prediction window is one year; longer horizons may show increased degradation. Fourth, gross drug cost basis differs from Medicare net liability: the universe gross total is approximately \$276 billion while CMS net Part D paid claims is approximately \$113 billion. Fifth, peer benchmarking uses state-level medians; specialty-adjusted baselines would refine the framework. Sixth, the Isolation Forest is one of several reasonable unsupervised anomaly detectors; comparison to alternative detectors (Local Outlier Factor, One-Class SVM, autoencoder anomaly scoring) is left for future work.

6. Conclusion

This paper presents a national-scale anomaly detection framework for Medicare Part D prescriber risk, with explicit EXCLDATE-filtered LEIE labels and temporal-persistence validation. On 1.38 million prescribers, the framework flags 69,034 providers (top 5 percent), with the flagged set showing 97.6 percent persistence into the next year and 8.59x lift on cost per beneficiary. The methodology corrects label-leakage problems common in prior supervised approaches. The framework, code, persisted result files, dashboard, and reproducibility envelope are publicly accessible.

Data and Code Availability

All source data is from CMS public-use files and the HHS-OIG LEIE database. The complete code repository, persisted result files, dashboard, and reproducibility envelope (RUN_INFO.json with SHA-256 hashes, library versions, and all parameters) are publicly available at [GitHub URL to be inserted upon submission].

Conflicts of Interest

The author declares no conflicts of interest.

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